

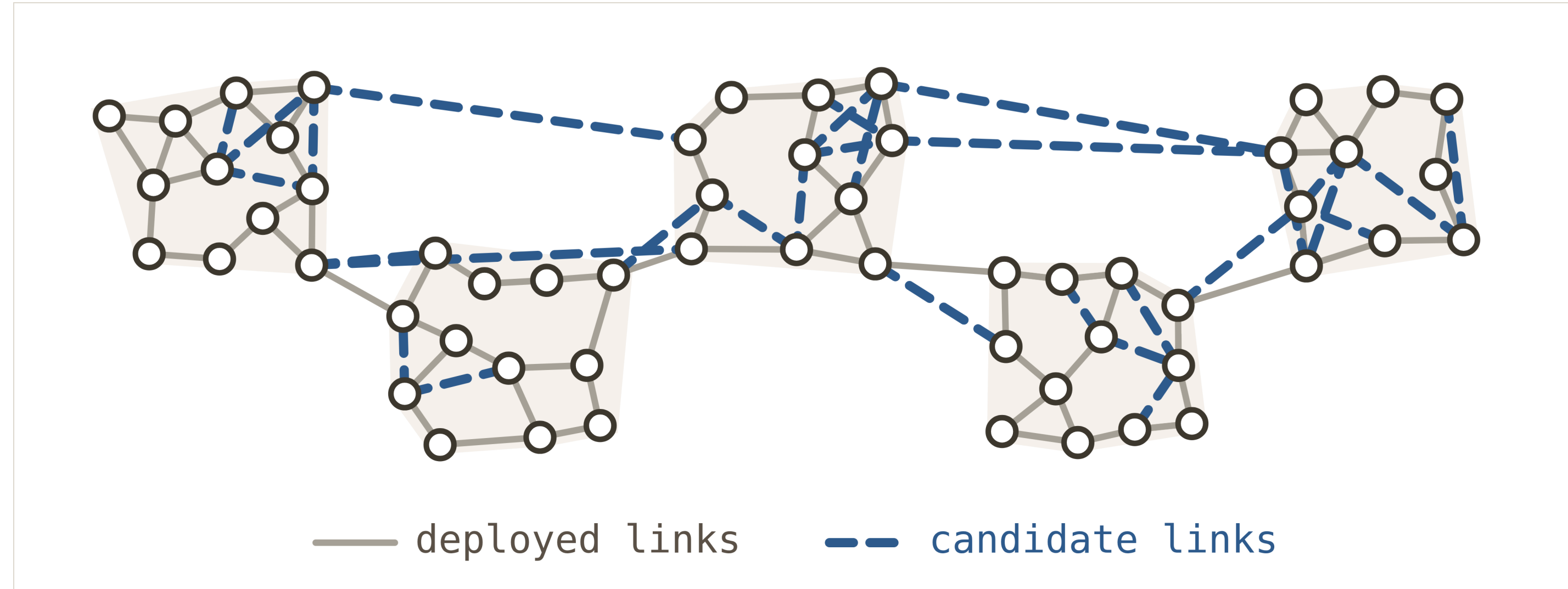
Cost-aware graph augmentation for robust networks via effective resistance

Omkar Bhoite[†] · V. Sateeshkrishna Dhuli[†] · Stefan Werner^{†,*} · Kimmo Kansanen[†]

[†] Department of Electronic Systems, Norwegian University of Science and Technology, Trondheim, Norway · ^{*} Department of Information and Communication Engineering, Aalto University, Espoo, Finland
{omkar.bhoite, venkata.s.dhuli, stefan.werner, kimmo.kansanen}@ntnu.no

01 Optimizing a pre-deployed network

Total effective resistance R_{tot} (the Kirchhoff index) is a tractable spectral proxy for robustness. Reducing it on a *deployed* weighted network means deciding **which links to add** and **how strong to make them** at once, one discrete choice coupled to a continuous one under a single global budget. Prices differ from candidate to candidate, combining a fixed per-link installation overhead with a penalty on the link's length.



The setup. Grey links and their weights are already deployed and **cannot be changed**. Dashed blue links form the candidate set $\mathcal{E}_c$; we choose exactly k of them and size each one, $w_i \in [w_{\min}, w_{\max}]$, at its own unit price c_i , spending no more than C in total.

OBJECTIVE

$$\mathbf{L}(\mathbf{w}) = \mathbf{L}_b + \sum_{e \in \mathcal{E}_c} w_e \mathbf{a}_e \mathbf{a}_e^\top$$

$$R_{\text{tot}}(\mathbf{w}) = n \operatorname{Tr} \left[\left(\mathbf{L}(\mathbf{w}) + \frac{\mathbf{1}\mathbf{1}^\top}{n} \right)^{-1} \right] - n$$

02 Joint selection and sizing

MIXED-INTEGER PROGRAM

$$\begin{aligned} & \underset{\mathbf{w}, \mathbf{z}}{\text{minimize}} && R_{\text{tot}}(\mathbf{w}) && \text{robustness of the upgraded network} \\ & \text{subject to} && \mathbf{c}^\top \mathbf{w} \leq C && \text{one global budget} \\ & && z_i w_{\min} \leq w_i \leq z_i w_{\max} && \text{perspective box, } z_i = 0 \Rightarrow w_i = 0 \\ & && \mathbf{1}^\top \mathbf{z} = k, \quad z_i \in \{0, 1\} && \text{exactly } k \text{ new links} \end{aligned}$$

($\mathcal{P}_0$)

03 A relaxation that certifies itself

Two things make ($\mathcal{P}_0$) intractable, the binary z_i and the matrix inverse inside R_{tot} . Relax $\mathbf{z}$ to $[0, 1]^p$; then, since $\mathbf{M}(\mathbf{w}) \succ 0$, bounding $\operatorname{Tr} \mathbf{M}^{-1}$ by $\operatorname{Tr} \mathbf{R}$ is *exactly* a Schur-complement condition. The inverse disappears, the problem becomes convex, and $\mathbf{M}$ is never factorized.

SEMIDEFINITE SURROGATE

$$\begin{aligned} & \underset{\mathbf{R}, \mathbf{w}, \mathbf{z}}{\text{minimize}} && n \operatorname{Tr}(\mathbf{R}) - n \\ & \text{subject to} && \begin{bmatrix} \mathbf{M}(\mathbf{w}) & \mathbf{I} \\ \mathbf{I} & \mathbf{R} \end{bmatrix} \succeq 0 \\ & && \mathbf{c}^\top \mathbf{w} \leq C \\ & && z_i w_{\min} \leq w_i \leq z_i w_{\max}, \quad i = 1, \dots, p \\ & && \mathbf{1}^\top \mathbf{z} = k, \quad 0 \leq z_i \leq 1, \quad \mathbf{R} \in \mathbb{S}_+^n \end{aligned}$$

($\mathcal{P}_1$)

Rounding and repair. Keep the k largest entries of $\mathbf{z}^*$, ties to cheaper links $\rightarrow$ project the weights onto the induced box $\rightarrow$ if over budget, scale toward $w_{\min} \hat{\mathbf{z}}$ until $\mathbf{c}^\top \hat{\mathbf{w}} \leq C$. The result is always a genuine, buildable design.

PROPOSITION 1 · INSTANCE-WISE GUARANTEE

At any optimum $\mathbf{R}^* = \mathbf{M}^{-1}(\mathbf{w}^*)$, and

$$R_{\text{tot}}^{\text{SDP}} \leq R_{\text{tot}}^{\text{MI}} \leq R_{\text{tot}}^{\text{RND}}$$

so the relaxation is a **certified lower bound** and every rounded design reports its own gap $(R_{\text{tot}}^{\text{RND}} - R_{\text{tot}}^{\text{SDP}}) / R_{\text{tot}}^{\text{SDP}}$.

04 Solved as a cone program

Stacking the unknowns as $\mathbf{x} = [\operatorname{vec}(\mathbf{R})^\top \mathbf{w}^\top \mathbf{z}^\top]^\top$ turns every constraint of ($\mathcal{P}_1$) into membership in a single product cone.

$$\mathcal{K} = \underbrace{\mathbb{S}_+^{2n}}_{\text{LMI}} \times \underbrace{\mathbb{R}_+}_{\text{budget}} \times \underbrace{\mathbb{R}_+^p \times \mathbb{R}_+^p \times \mathbb{R}_+^p \times \mathbb{R}_+^p}_{\text{perspective box and } z \text{ bounds}} \times \underbrace{\{0\}}_{\text{cardinality}}$$

- Solved by ADMM on the homogeneous self-dual embedding (SCS).
- A **cached LDL^T factorization** reduces each affine step to two triangular solves.
- One eigendecomposition of the PSD block then dominates each iteration.

05 Greedy baseline with rank-one updates

Sherman and Morrison turn each candidate evaluation into a closed-form marginal gain, so the greedy adds links sequentially without ever re-inverting $\mathbf{M}$. A **tentative budget allocation** $\mathbf{s} = C_{\text{rem}}/k_{\text{rem}}$, the budget left over the links left, prices each trial weight.

MARGINAL GAIN, EXACT

$$\Delta R_{\text{tot}} = \frac{n w_{(v_i, v_j)} b^2(v_i, v_j)}{1 + w_{(v_i, v_j)} r(v_i, v_j)}$$

$$\mathbf{r} = \mathbf{a}^\top \mathbf{M}^{-1} \mathbf{a}$$

$$b^2 = \|\mathbf{M}^{-1} \mathbf{a}\|_2^2$$

TRIAL WEIGHT

$$\mathbf{w}_e = \max\{w_{\min}, \min\{w_{\max}, s/c_e\}\}$$

from the tentative allocation $\mathbf{s} = C_{\text{rem}}/k_{\text{rem}}$

$$\text{total cost } \mathcal{O}(n^3 + k(pn + n^2))$$

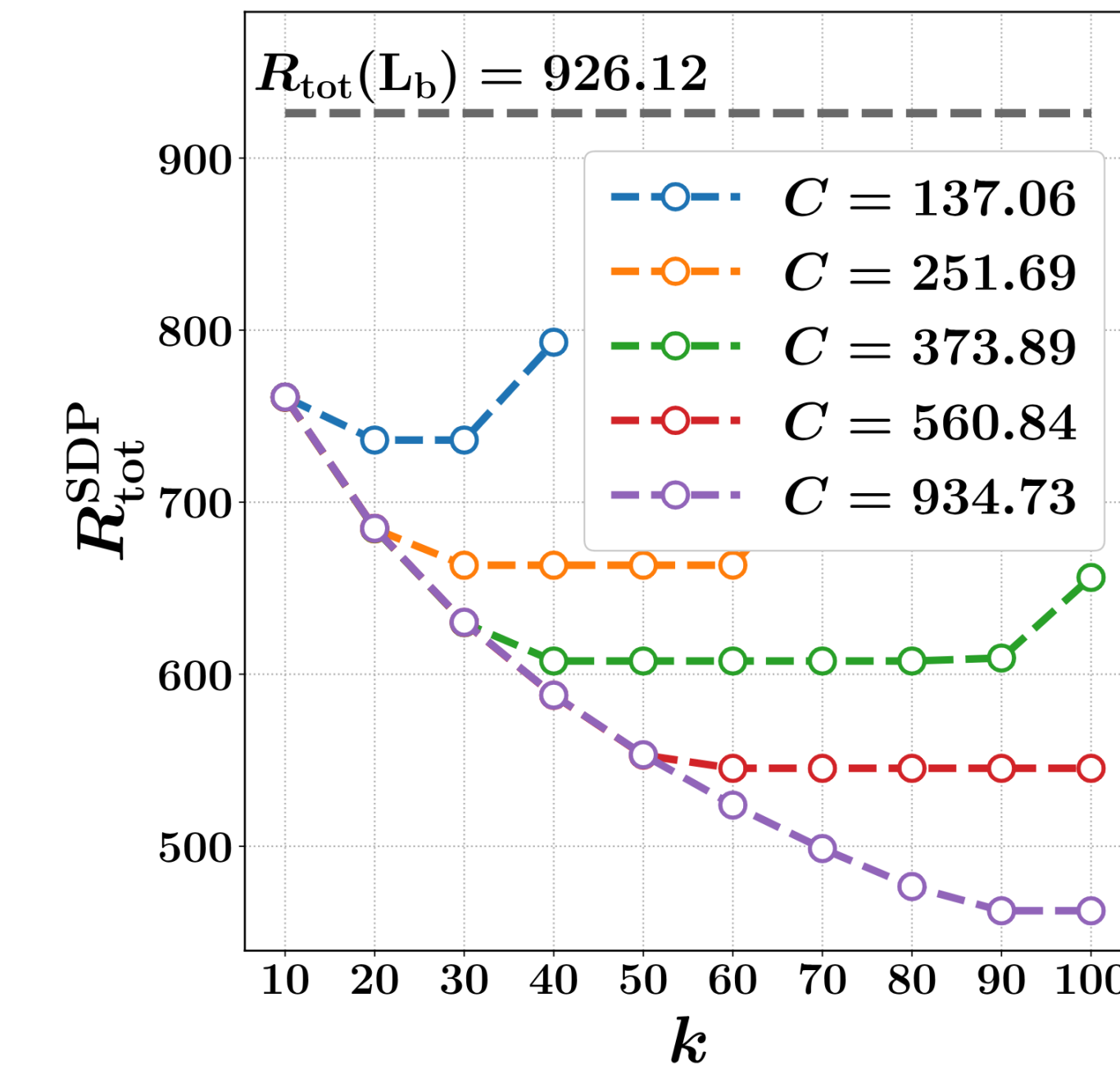
06 Instances and candidate sets

Synthetic. Planar base graphs with lognormal weights. Candidates are the intra-region k -NN links of a k -means partition plus a sparse inter-region backbone.

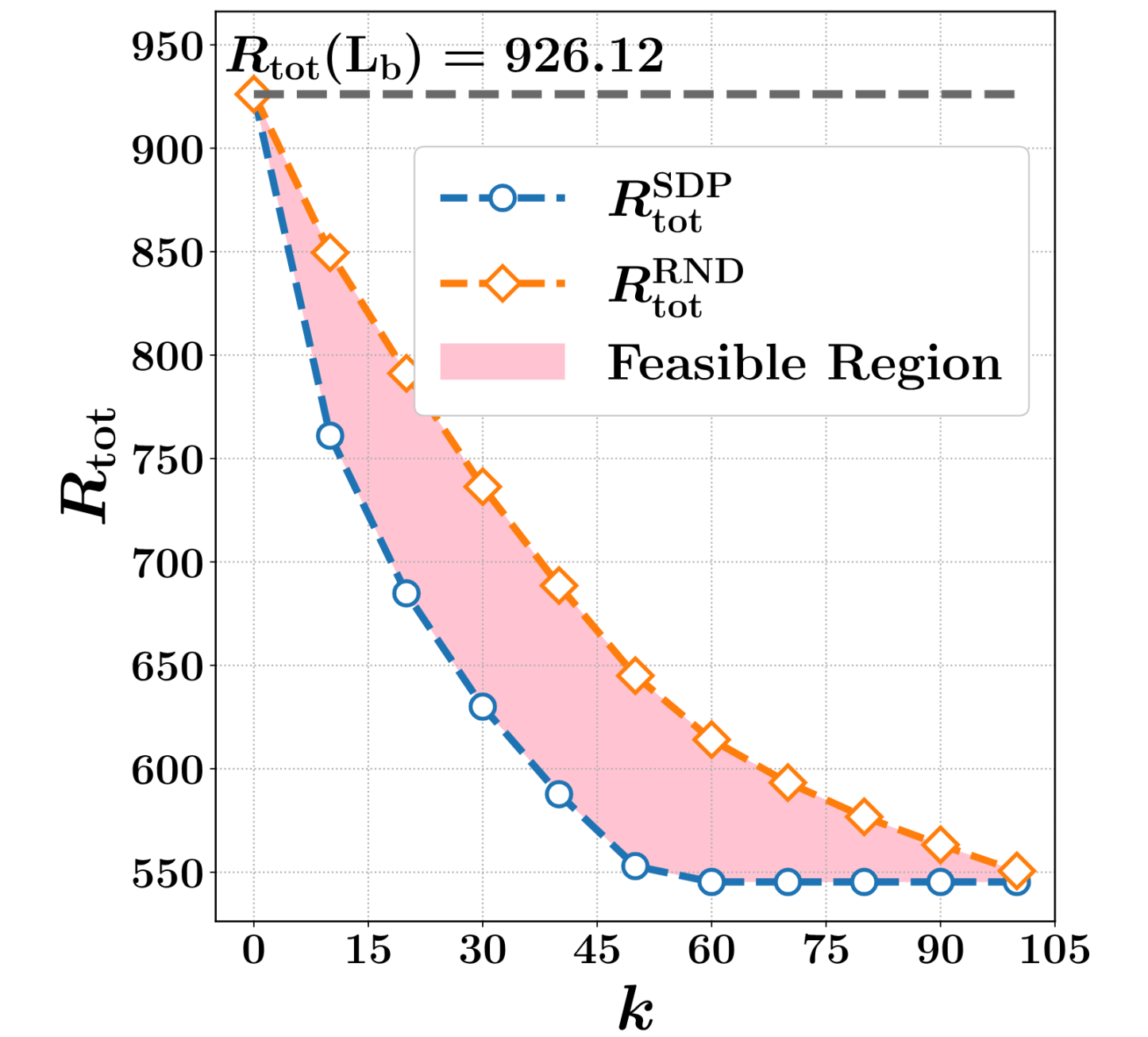
Real-world. A connected subgraph of the California road network, same candidate construction, priced by $\mathbf{c} = c_{\text{fixed}} \mathbf{1} + \beta \mathbf{d}_c$.

07 What the budget buys

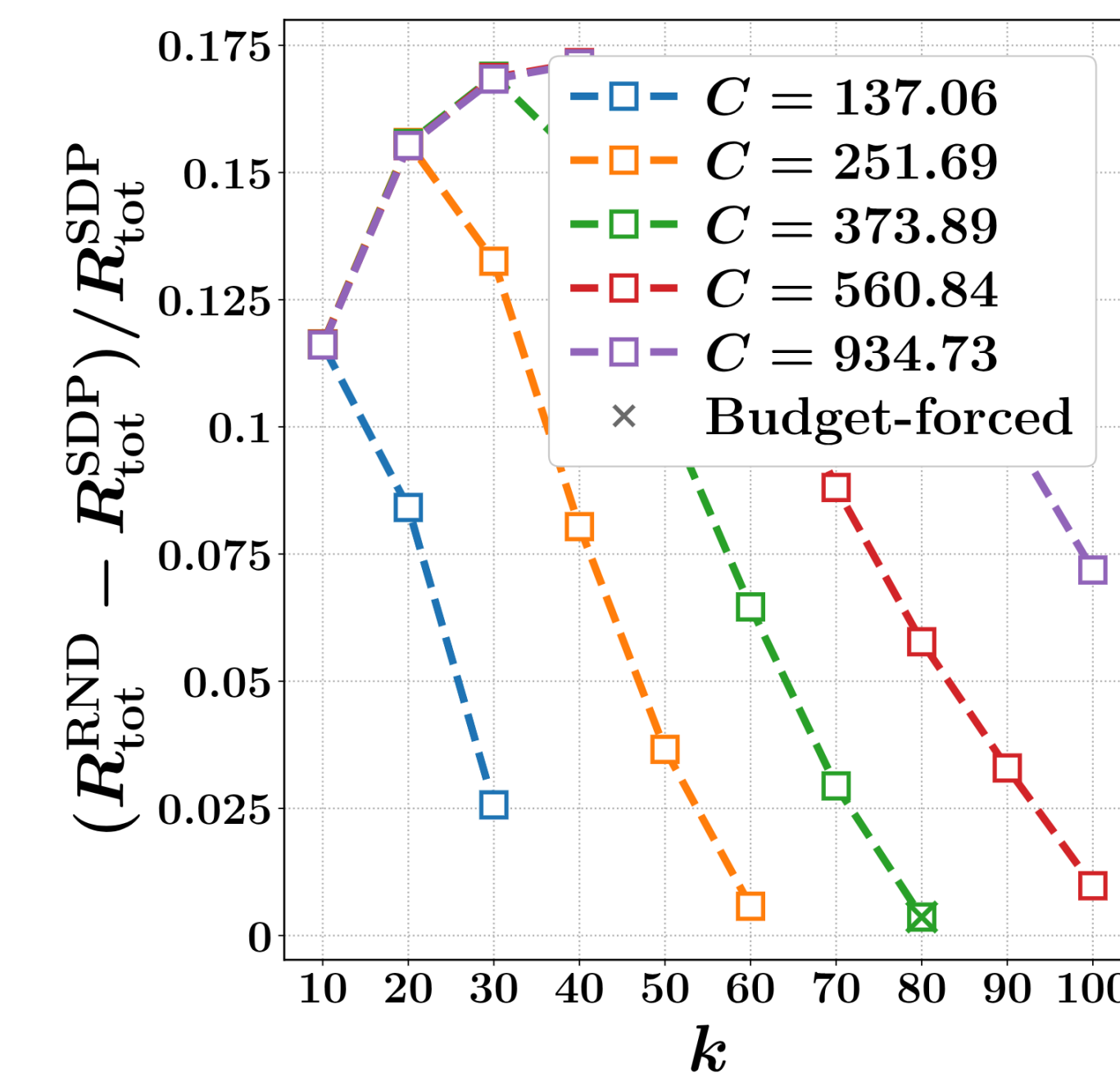
From a base $R_{\text{tot}}(\mathbf{L}_b) = 926.12$ the certified bound falls fast, then saturates. Budgets are deduced from the candidate cost structure, so the sweep spans binding, marginal and slack regimes.



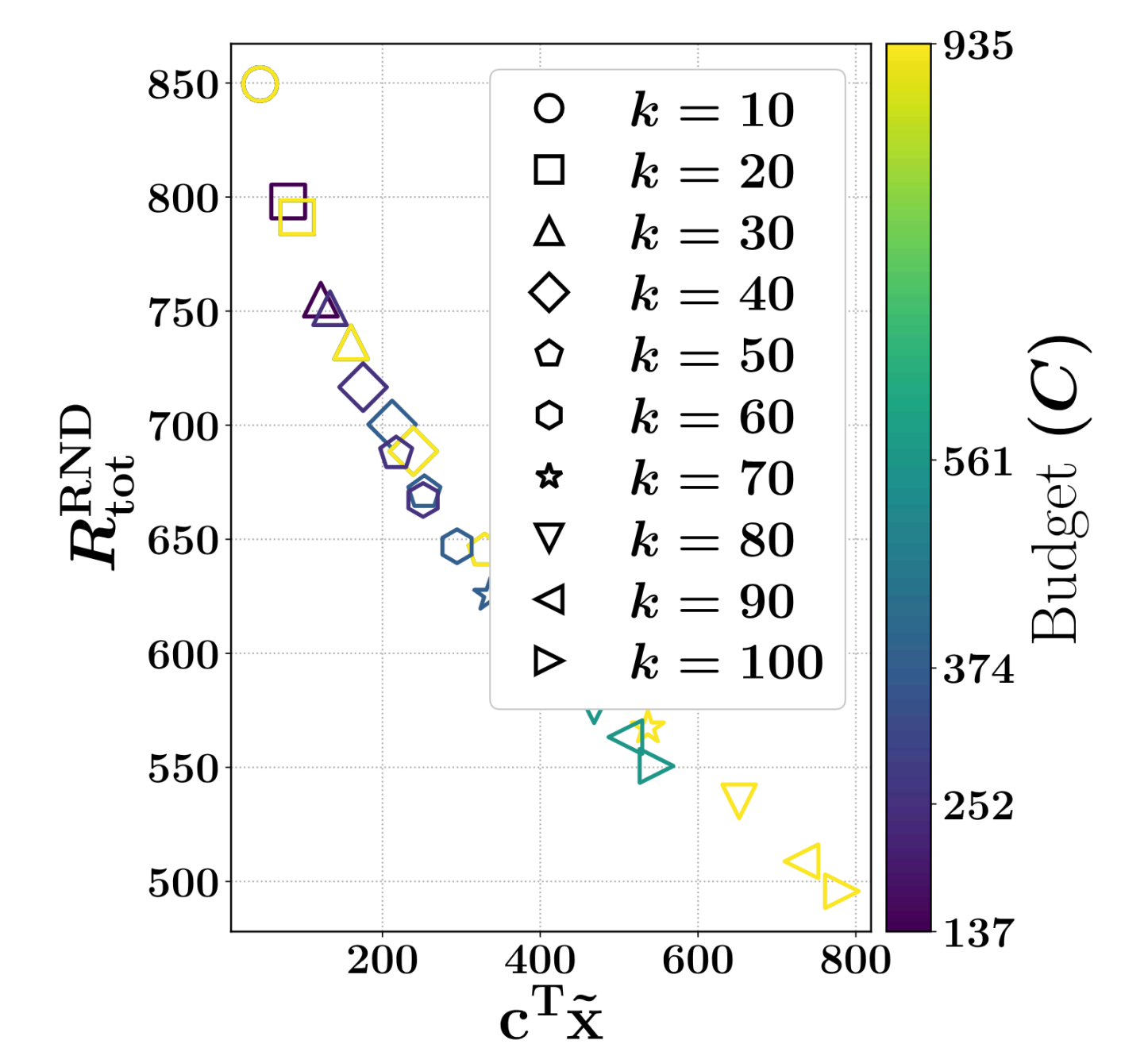
1a. Certified bound versus k at five budgets.



1b. Feasible region, bound to rounded design.

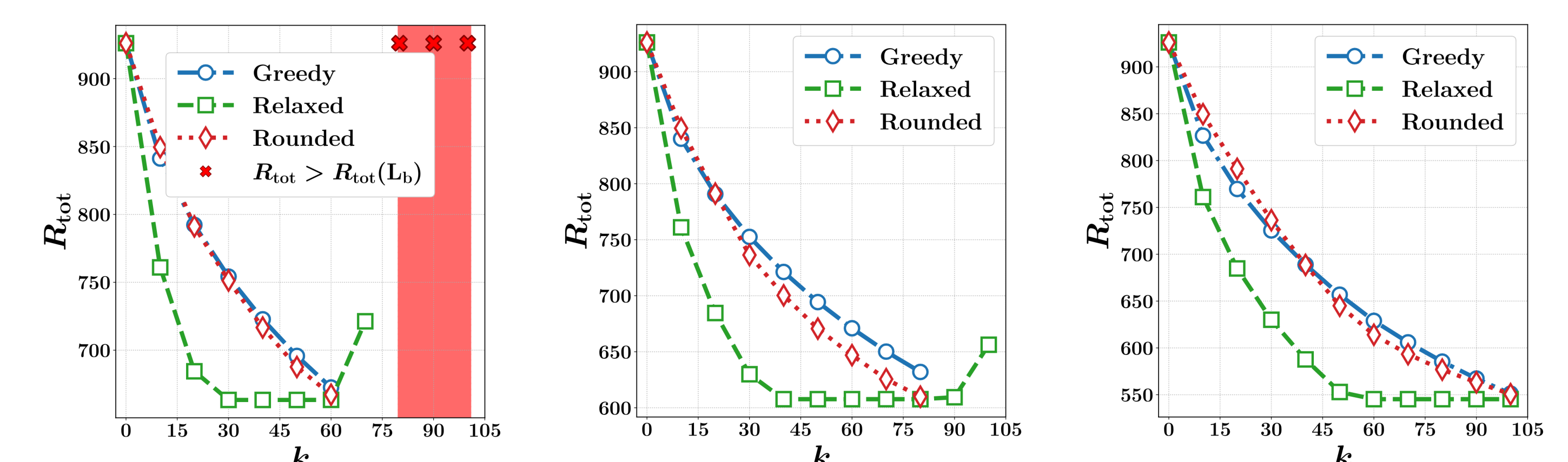


1c. Relative gap $(R_{\text{tot}}^{\text{RND}} - R_{\text{tot}}^{\text{SDP}}) / R_{\text{tot}}^{\text{SDP}}$.



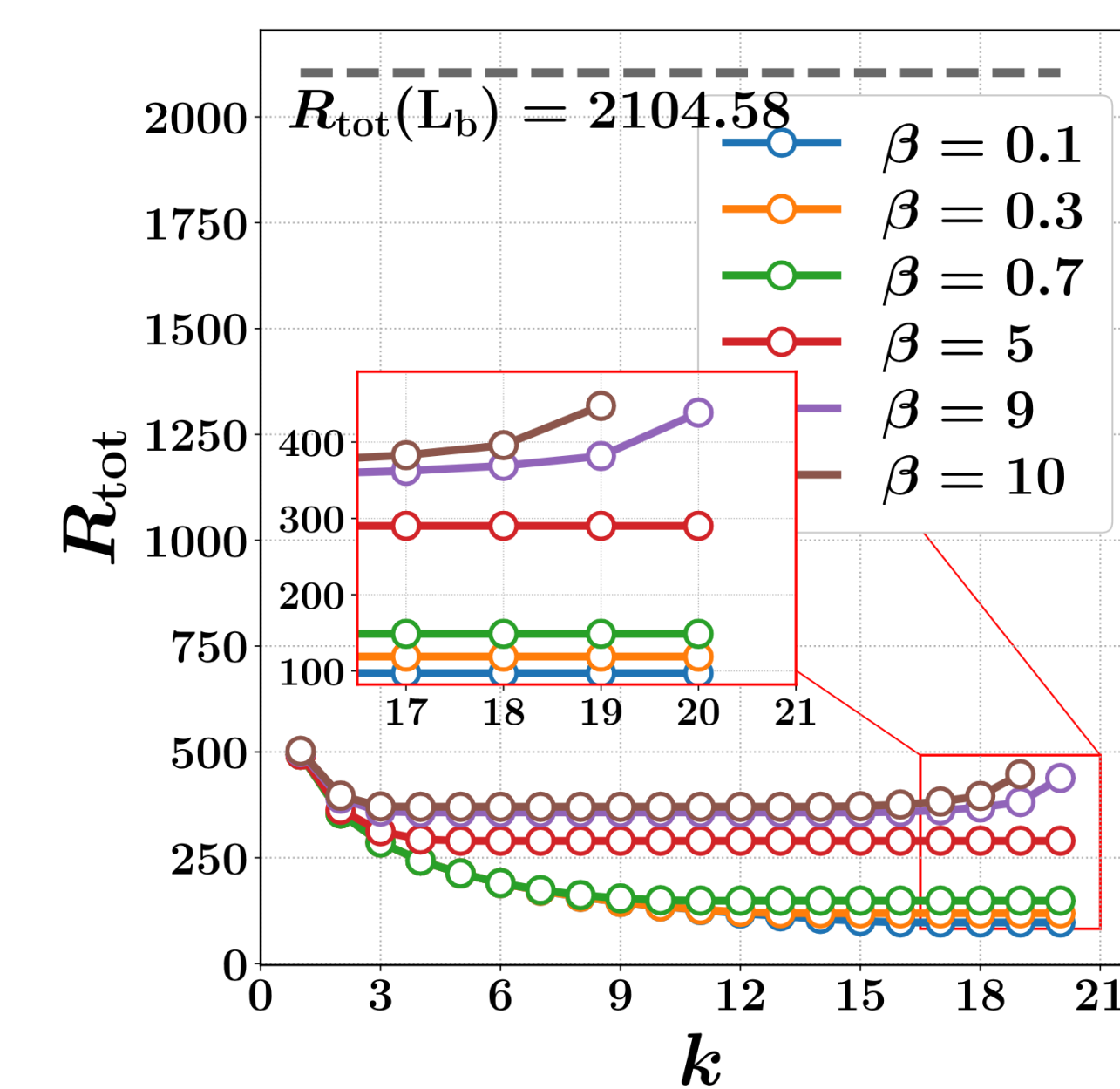
1d. Pareto front against budget spent, fixed k .

08 Three budget regimes



Constrained. $C = 251.69$. Infeasible from $k = 80$; greedy stalls after 61 links. **Marginal.** $C = 373.89$, the anchor. Feasible, but on the boundary. **Relaxed.** $C = 560.84$. All k feasible; rounded tracks the bound.

09 California road network



2a. $n = 500$ nodes, $C = 150$, sweeping β .

2104.58

base $R_{\text{tot}}(\mathbf{L}_b)$ of the sampled subgraph, every deployed road weight set to one

$w_i \in [1, 5]$

the range each new road link may be sized in

20 edges

enough to make ($\mathcal{P}_1$) infeasible once $\beta = 10$

10 Takeaways

- Augmenting a *deployed* network is not the same problem as designing one. The base Laplacian reshapes both the objective and the cost trade-off.
- The SDP relaxation is a **certificate**, not just a heuristic. Every discrete design ships with its own optimality gap.
- Budget, not cardinality, is the binding resource. Near the anchor budget, added links are forced toward $w_{\min}$ and buy almost nothing.
-



Paper & code
omkarbhoite25.github.io